GoServer

N-player Go in the spirit of KGS

Project team

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Contents

1 What GoServer is **1**
- 1.1 Running it . . . 1

2 Rooms, challenges and chat **2**
- 2.1 Navigation . . . 2
- 2.2 Identity . . . 2
- 2.3 Bounds . . . 2
- 2.4 Chat follows the tree . . . 3
- 2.5 Correlation and replay . . . 3

3 N-player Go rules **4**
- 3.1 Board, turns and captures . . . 4
- 3.2 Losses, bowls and score . . . 4
- 3.3 Ko and repetition . . . 4
- 3.4 Clocks and ending . . . 4
- 3.5 Challenge parameters . . . 5

4 Seats, bots and teams **6**
- 4.1 Seat plan . . . 6
- 4.2 Bots . . . 6
- 4.3 Autopilot . . . 6
- 4.4 Team merges . . . 7

5 Claimed names and ranking **8**
- 5.1 The sign-in form . . . 8
- 5.2 Which games count . . . 8
- 5.3 The rating . . . 9
- 5.4 What survives a restart . . . 9

6 What every viewer sees **10**

7 Requirements **12**
- 7.1 Shape . . . 12
- 7.2 Positive invariants . . . 12
- 7.3 Negative requirements and their controls . . . 12
- 7.4 Derived outputs . . . 14

8 Implementation **15**
8.1 Evidence 15

9 Usage scenarios **16**
9.1 Welcome 16
9.2 Creating a challenge 17
9.3 Planning seats 18
9.4 Aiming a stone 19
9.5 Reading the bot's search 20
9.6 A watcher explores a variation 21
9.7 Proposing a team merge 22
9.8 Playing as a team, on autopilot 23
9.9 The result 24
9.10 Reviewing together 25
9.11 Rooms and games as tabs 26

10 Known gaps **27**

A Task catalog **28**

B How the project got here **36**

Chapter 1

What GoServer is

GoServer is a local, in-memory online Go service for *multiplayer* Go: two to eight players on one board, each with a colour, taking turns in order. KGS inspires the navigation (rooms, challenges, chat) but there is no compatibility with KGS accounts, protocols or clients. The rules engine is a port of the stakeholder's single-page N-Player Go Lab, whose JavaScript (`source/cmd/demo/static/engine.js`) remains the reference.

A visitor picks a display name, enters a room, talks in room chat and opens or creates a challenge. The creator chooses the board and clocks, decides seat by seat whether a person or a bot plays and who is on which team, and starts the game. Empty seats become bots that search with Monte Carlo tree search on the server. During play a person can hand their seat to a bot (autopilot) and take it back, and teams can merge when everyone on both teams agrees. Every viewer can step through the game's history and see each bot move's search; watchers can branch and run their own searches, which never reach the server.

The record of the project is this book together with `spec/Requirements.lean`, `spec/Requirements.proto`, `spec/Implementation.lean` and the source in `source/`. Review and acceptance happen in git; chat histories are disposable.

1.1 Running it

From the project root:

- `make run` builds and serves the app on `127.0.0.1:8081` (`make run PORT=8090` picks another port; LeanFM's own 8080 is never used); `make stop` stops only the process it started.
- `make spec-check` runs LeanFM's canonical check on the three specification files and regenerates `spec/tasks.json`.
- `make views` derives the task figures of Appendix A from that catalog.
- `make app-check` builds and runs `go test -race`.
- `make browser-check` drives a real headless Chrome through two users' room, challenge and chat flow; `make tour` replays the usage scenarios of Chapter 9 and refreshes their screenshots.
- `make book` rebuilds this book in light and dark copies.

All state lives in memory and resets on restart. Building needs Go and Node.js with npm: the browser app is a React project in `source/web` that `make run` builds first (the first build runs `npm ci`).

Chapter 2

Rooms, challenges and chat

2.1 Navigation

After choosing a name a person is in the lobby. Every room they have open is a tab along the top, and a final *+ Rooms* tab lists all rooms and creates new ones. Inside a room, the first tab is its Lobby: the room conversation, its challenges and the form for a new challenge. Opening a challenge adds a tab for that game, so several games can be open at once, in one room or in several. A game's tab shows its seats, its board once started, and its own chat. Tabs stay alive in the background, keeping each game's variations and analysis; a room tab counts unread messages, and a game tab is marked when a move or a merge answer is waiting for the viewer. Open tabs come back after reloading the page. Each tab shows only results requested for its own room or game, so a slow response can never appear in another tab. Room chat and game chat are separate conversations: nothing posted in one is copied to the other, and only members of a room or game receive its chat. Watching or joining a game enters its chat; neither grants any move authority.

2.2 Identity

The server issues the session; creator and sender are always derived from it, never taken from request fields. A name is free until its holder claims it with a password (chapter 5); an unclaimed display name identifies a person for the life of the server, as before. Sessions live in memory, so restarting the server forgets them; a page that still holds a forgotten session returns to the name form, prefilled with the previous name, and says why.

2.3 Bounds

Names are 1–128 Unicode characters, trimmed; room names are unique and case-sensitive and are never interpreted as paths or markup. Chat bodies are 1–4000 characters; each room keeps its latest 100 entries and each game its latest 500. There are at most 16 rooms including the lobby and 64 challenges in total. Each session has a finite budget of admitted mutations; once it is spent, new mutations are rejected while already accepted results stay replayable, so a retried request never applies twice.

2.4 Chat follows the tree

Every game chat message is pinned to the position the sender was looking at: a position of the game (at most the moves played so far) or, after the finish, a node of the review tree. Server notices are pinned to the position they report on. The Chat tab shows the remarks pinned to the positions on the line from the empty board to the position on the board, in the order of that line, each tagged with its move (or *here*). So a player can step back four moves and write "this was the losing move": stepping back further hides it, and anyone who later walks to that position, along the game or into a variation, sees it.

2.5 Correlation and replay

Every request carries a request ID unique per session and task, and receives exactly one result: its success message or `RequestRejected`, with the same ID. A duplicate mutation returns the earlier result; reusing an ID with a different body is rejected. A rejected or overloaded request changes nothing and discloses no other person's identity or session.

Chapter 3

N-player Go rules

3.1 Board, turns and captures

The board has 5, 7, 9, 11, 13 or 19 lines and is bounded by a Wall. Seats move in order. After a stone is placed, every adjacent group of *another* colour left without liberties is removed first. Only if nothing was captured and the placed stone's own group has no liberties is the move a suicide, and then that whole group is removed. Suicide is legal: in multiplayer Go a player may sacrifice a group to help an ally make eyes.

3.2 Losses, bowls and score

A removed stone counts against its *owner*, not for the capturer: it goes to the owner's losses, and the bowl records which seat captured it (suicide is recorded against the mover). Territory is an empty region bordered only by one colour's stones and the Wall. A seat's score is its territory minus its losses; a team's score is the sum of its members' territory minus the sum of their losses,

$$S_T = \sum_{p \in T} A_p - \sum_{p \in T} G_p.$$

The winners are the team or teams with the highest score among teams that still have a member who has not resigned.

3.3 Ko and repetition

The ko rule is a challenge parameter: *Auto* (the default: on for two players, off for three or more), *Off*, or *On*. When on it is simple ko plus *situational* superko: a stone may not recreate an earlier board with the same seat to move. The Lab calls this "positional superko", but its comparison includes the player to move, so situational is the accurate name. A Zobrist hash only indexes candidates; equality is decided on the exact board and seat, so a hash collision can never reject a legal move. Passes, timeouts and resignations are never rejected for repeating a position.

3.4 Clocks and ending

Every seat, person or bot, plays on the same Japanese byo-yomi clock: main time (1, 5, 10, 20, 30 or 60 minutes; default 5), then a number of periods (1, 3, 5 or 10; default 5) of equal length

(10, 20, 30 or 60 seconds; default 30). A turn may use what is left of the main time and then the periods. A move made within a period keeps that period; every period run over is used up. When the last period runs over, the turn is a *timeout*: it passes the turn but neither ends nor resets a round of passes, and the seat keeps one period for its later turns. A move arriving after the seat's clock ran out is rejected and the timeout wins. Switching autopilot on or off never touches the clock. The game ends when every remaining seat has passed since the last stone, after one full round of timeouts (scored as it stands), or at $3N^2$ moves on an $N \times N$ board. A resigned seat passes automatically and cannot win. The final scores and winners are announced in game chat.

3.5 Challenge parameters

Board size (default 11), seats 2–8 (default 4), the byo-yomi clock, the ko rule, and the bot settings of the Lab: exploration constant 0.5, 0.7 (default), 1.0 or 1.4; reward *margin* (a soft margin against the best rival, the default), *win* or *rank*; and margin softness 2, 4 (default) or 8 points. Zero or empty means the default; anything out of range is rejected.

Chapter 4

Seats, bots and teams

4.1 Seat plan

Before the start the creator plans each seat as *bot* or *open* and gives it a team number from 1 to the seat count; by default each seat is its own team. People in the room take open seats (one seat per person per game) or leave them. A seat planned as a bot cannot be taken, and a seat a person holds cannot be turned into a bot seat until that person leaves. Only the creator starts the game; every seat still without a person becomes a bot named `bot-`*seat*. Seats never change after the start.

4.2 Bots

Bots run Monte Carlo tree search on the server, for their team's combined score. A bot turn is dispatched to a bounded pool of search workers (at most $\min(8, \mathit{CPUs}/2)$ at once), each with a time budget and a 200,000-node cap, and the search never holds the game lock. A result is applied only to the game, seat and turn revision it was dispatched for, and only before the seat's clock runs out; a late result is discarded (Figure A.25). No client can submit a bot's move, and a bot never moves for a person's seat.

A bot plays on its seat's clock and manages it so that it has plenty of time but does not waste it on easy moves. In main time it plans a share of what is left, divided by the moves it still expects to play (at least eight, and the empty points over the seats); in byo-yomi it plans half a period. It may stretch a close call to four shares, or into most of one period once main time runs short, but it never plans to use up a period. While it searches it checks how fast the playouts come: once the second-best move could not overtake the best in the time it has left, the choice is *settled* and it stops at once; at the planned time it stops unless the top two moves are close (the second has at least 70% of the best's visits); a move with one candidate takes no time. Each bot move publishes its search: playouts, time, the time it planned and why it stopped, the top candidates with visit counts, the mover's expected reward, and every seat's projected final score and outlook. The status line says, for example, "thought 2.2 s of a planned 2.2 s (stopped at the plan)".

4.3 Autopilot

A seated person may switch autopilot on, letting a server bot play their seat on the seat's own clock with published search, and switch it off at any time. The seat, team and name stay theirs. Switching off during their own turn discards the pending bot result and leaves them whatever the seat's clock still holds for that turn; neither switch adds or removes time.

4.4 Team merges

During play a seated person may propose merging their team with another team that also has a seated person (nobody could consent for an all-bot team). The merge happens once every person seated in either team accepts; it takes the smaller team number and is permanent. Any decline, any change to either team, or the game finishing withdraws the proposal; there is at most one pending proposal per pair of teams. Bots of the merged team then search for its combined score (Figure A.27).

Chapter 5

Claimed names and ranking

5.1 The sign-in form

Everything about a name happens on the sign-in form, and the form reads what you filled in:

- **A name alone**: you play as a guest under that name, as long as nobody has claimed it and nobody else is signed in with it.
- **A name and one password**: you sign in to your claimed name. Any older session for the name ends; your rooms, games and seats stay, because they belong to the name.
- **A name and the same new password twice**: you claim the name and are signed in. This works only if nobody has claimed it and nobody is signed in with it as a guest. If you were that guest, press *Sign out* first; your seats come with you.

Below the form, *Manage a claimed name* opens a panel that is closed by default. *Change password* takes the current password once and the new one twice; it signs out every other session for the name. *Release name* takes the password and a ticked confirmation; the name becomes free for anyone and leaves the ranking. Its rated games keep counting for the other players, but a later claim of the same name starts afresh, because each claim is a different player to the rating.

Claims ignore letter case and surrounding spaces, so `Rob` and `rob` are one name. Names beginning `bot-` belong to bots and nobody can sign in with one.

The server stores a salted PBKDF2-SHA256 hash (600,000 rounds), never the password. After five wrong passwords in a row for one name, each further attempt (to sign in, change the password or release the name) waits twice as long as the last (2 seconds, then 4, up to 15 minutes), and only one attempt per name is checked at a time; at most two hashes are computed at once. A name changes its password at most once a minute. There is no password reset: a forgotten password means a new name.

5.2 Which games count

A finished game is *rated* when seats on at least two different teams are held by different rated players. A bot is rated by its search settings (time, exploration, reward), so a 4-second bot and a 1-second bot are different players and beating the stronger one counts for more. A person is rated only if their exact name was claimed before the game started and their seat never switched to autopilot. Other seats still play; they just do not feed the ranking. Games before a claim never count, because anyone could have used the name then.

5.3 The rating

Teams are placed by whether they still have a member who has not resigned, then by team score; equal teams share a place. Every two rated seats on different teams make one comparison: a win and a loss, or half of each when the teams tie. Each pair of teams carries a total weight of one, shared among their rated members, so a team of three does not outvote a lone player.

The ratings are the Bradley–Terry maximum-likelihood fit over all comparisons, on the Elo scale (400 points per factor of ten in strength). Every player also gets one win and one loss against a fixed 1500 anchor, which keeps ratings finite and holds players with few games near 1500. The fit uses every recorded game at once, so the ranking never depends on the order in which games finished, and recomputing it from the file always gives the same numbers: the weights are exact integers, so even the floating-point sums do not depend on order. Some values from the tests:

- One game, A beats B: A 1592, B 1408 (checked by hand).
- One four-player game in places 1 to 4: 1713, 1566, 1434, 1287.
- A team of A and B beats C: A and B 1551 each, C 1396.

The Ranking tab lists claimed names and bots by rating, with each one's rated games and first places; it refreshes every ten seconds.

5.4 What survives a restart

Claims and rated results are appended to `data/goserver.jsonl` in the project folder (`GOSERVER_DATA_DIR` overrides it). Each record is one line, flushed to disk before the claim is acknowledged. On start the file is replayed: a line cut off by a crash is dropped, and any other damaged line stops the start with its line number rather than silently losing history. The folder is private to the user (mode 0700, file 0600), symlinks are refused, and no web route reads it. Up to 1000 claims at once, 20,000 account records (claims, password changes and releases) and 100,000 rated games are kept; beyond that a game says in its chat that it was not added. Rooms, games, chat and sessions still reset on restart.

Chapter 6

What every viewer sees

The browser reproduces the Lab against server state:

- the board with Wall coordinates, territory squares, the last-move marker, ghost markers for next moves and a search heat overlay;
- a translucent stone in the mover's colour under the pointer wherever a click would place a stone (the viewer's own live turn, or a variation where analysis is open to them); over an occupied or forbidden point, or out of turn, there is no preview and the pointer shows that a click would be refused;
- player colours drawn as stones on a small wood tile, so a black stone never reads as a hollow white ring on the dark page;
- an always-visible play bar (whose turn it is, the clock showing main time or "byo-yomi 0:23 × 3", Pass, Resign, autopilot) above tabs for Scores, Search, Seats, Chat and GIF; the Chat tab counts unread messages and the Seats tab flags a merge awaiting the viewer's answer;
- navigation with Home, Back, Forward and End, by buttons and by the arrow, Home and End keys; where a position has two or more next moves (labelled A, B, ... on the board and in the next-move chips), up and down choose which one Forward follows, without moving; elsewhere they switch to the previous or next sibling variation; a variation tree;
- a *Search at this position* panel listing candidate moves with visit share and reward, from the bot's published search or the viewer's own;
- per seat: Territory, Bowl, Score, Proj. (projected final score), Outlook (expected reward) and Clock (main time left, or periods left), and team totals;
- a clickable projected-score timeline along the viewed line;
- pending merge proposals with who has accepted and who must still answer;
- GIF export of the viewed line, and stone, capture and last-ten-seconds sounds with a toggle;
- live following while at the latest move, otherwise a notice that new moves arrived;
- under the board, a *Session type of GoServer* panel that stays closed until opened: the sequence diagram LeanFM draws from `spec/Session.lean`, the session-type text it was drawn from, and what the picture answers (question, source, reducer, unit and boundary).

The panel's files are generated by `spec/SessionPanel.lean` with LeanFM's own renderer, never drawn by hand; after a change to `spec/Session.lean`, run `lake env lean --run projects/GoServer/spec/Ses` from the LeanFM root, or `TestSessionPanelMatchesTheSpec` fails. During play, variations and local searches run in the viewer's browser with the Lab engine on a copy of the position. They are never sent to the server, never change the game, and vanish when the page closes. Once the game has finished, every variation move anyone in the game plays becomes part of a shared *review tree*: the server checks it against the rules from the position it follows, keeps it for as long as the game exists, and every game member sees it appear in their own tree; reloading the page loses nothing. The same move after the same position is one node. The review never changes the game, its result or the ranking, and local searches still stay in the browser. Watchers may branch and search at any time; a person seated in a game in play may browse the actual moves, and may branch and search once the game has finished. Wherever analysis is open, the viewer plays for whichever seat is to move in the position on screen: a click on a point, a pass with `P` or the Pass button, or the computer's choice with `C`; each becomes a variation. That last rule is a fairness convention of the provided browser, not a security boundary: a public position can always be copied into another tool. The page is dark-only, and its text is always light on a dark background: highlighted buttons and tags use yellow text and a yellow border on a dark fill, never dark text on yellow, so a browser's dark-mode extension cannot turn them into white on yellow.

Chapter 7

Requirements

7.1 Shape

`spec/Requirements.lean` declares two actors that people see, `Person` and `Server`, and a third, `Bot`: the bounded pool of server search workers. Every operation is a task whose grammar is a request followed by its correlated success or `RequestRejected`. Five tasks are composed: `bot_turn`, `turn` (a person's move, a bot turn, a timeout or an automatic pass), `team_merge`, and `game_lifecycle`: create, then any number of seat, join, chat and state operations, then start, then turns, autopilot switches, merges, chats and state reads, until `GameFinished` (Figure A.28). Server-originated events (`BotMovePlayed`, `TurnTimedOut`, `AutoPassed`, `TeamsMerged`, `MergeWithdrawn`, `GameFinished`) reach game members as entries of `GameState`, and `GetGameState` takes `since_move` so a viewer fetches only new move records. `spec/Requirements.proto` declares the same messages.

7.2 Positive invariants

Every room, chat, board, clock, score and team a viewer sees comes from one server-authoritative transition in a scope the viewer is authorized for. Every turn ends exactly once ($AG(\mathit{to_move} \Rightarrow AF\ \mathit{moved})$), and every started game finishes ($AG(\mathit{playing} \Rightarrow AF\ \mathit{finished})$).

7.3 Negative requirements and their controls

Each *never* obligation of the specification is held by a control and a test; `source/traceability.json` lists them in full.

- **A move for another seat, out of turn, or for an autopilot seat**: seat-to-move identity check under the game lock (`TestMovesAreAuthorizedOrderedAndRuleChecked`).
- **Two transitions for one turn** (move, bot result, timeout, auto-pass): a turn sequence number under the lock and the replay cache (`TestTurnsNeverEndTwice`).
- **A move accepted after its deadline**: the seat's clock is rechecked on arrival and the timeout applied first (`TestLateMoveLosesToTheDeadline`).
- **A clock that cheats**: periods used up only when run over, one timer per turn at the clock's end, autopilot switches that never touch it, a bot plan that never reaches the end

of a period, searches cancelled at shutdown (`TestByoyomiChargesMainTimeThenPeriods`, `TestMovesAreChargedAndAutopilotLeavesTheClock`, `TestBotPlanNeverUsesUpAPeriod`, `TestSearchStop`

- **A review move that changes the game, or that should not exist**: review moves only after the finish, only by game members, only after existing positions, checked by replaying the rules, capped at 2000 per game, kept apart from the record (`TestReviewTreeAfterTheFinish`).
- **A remark pinned to a position that does not exist**: the server resolves every pin and falls back to the latest actual position (`TestReviewTreeAfterTheFinish`; the usage tour checks the Chat tab follows the line and survives a reload).
- **A session-type picture that is stale or hand-drawn**: generated from the spec by LeanFM, checked against it, shown as an image (`TestSessionPanelMatchesTheSpec`, `browser_check.py`).
- **A stale or misdirected bot result**: results bound to game, seat and turn revision (`TestBotNeverMovesFor`, `TestAllBotGamePlaysToTheEnd`).
- **A legal move rejected by a hash collision**: exact situation compared behind the Zobrist key (`TestSituationalSuperkoUsesExactPositions`).
- **Unbounded bot work**: search outside the lock, time budget, node cap, bounded slots (race-enabled `TestAllBotGamePlaysToTheEnd`).
- **A merge without consent**: per-person acceptance sets and withdrawal on any team change (`TestMergesNeedEveryPersonsConsent`, `TestTeamChangeAndFinishWithdrawProposals`).
- **Seat or plan changes by the wrong person or after the start**: creator and status checks (`TestSeatPlanIsTheCreatorsBeforeStart`, `TestSeatsStartAndBotsFillEmptySeats`).
- **Crossing a room or game boundary**: membership checks on every read and write (`TestNamedChallengesAndChatHistoryStayInScope`).
- **Executable user text, fabricated destinations, stale responses replacing another scope, exposed tokens**: text-only DOM rendering, selection-generation guards, strict bounded JSON, Host and Origin checks (`TestMethodOriginHostAndBodyRejections`, `browser_check.py`).
- **A pointer preview that promises a refused move, or talks to the server**: the preview is gated by the same predicates as a click and drawn only on the canvas (`TestViewerTabsAndPointerPreview`; the usage tour hovers an empty and an occupied point in Chrome).
- **A claimed name used without its password**, including case variants and bot names: claim lookup by lower-cased name, constant-time hash comparison, the per-name doubling delay (`TestClaimFromTheSignInForm`, `TestWrongPasswordsAreThrottled`, `TestClaimSignInAndRankingOve`, `browser_check.py`).
- **A password kept or shown**: only a salted hash is stored; the claim never enters the replay cache; responses, the page, its storage and the data file are checked for it (`TestClaimFromTheSignInForm`, `TestDataFileIsDurableAndRefusesDamage`, `browser_check.py`); PBKDF2 is checked against RFC 7914 (`TestPBKDF2MatchesRFC7914Vector`).

- **A rating from a game that should not count**, or a game counted twice or depending on order: claim time against game start, the autopilot flag, random result identities, exact integer weights (`TestFinishedGamesFeedTheRanking`, `TestFitMatchesIndependentNewtonFit`, checked against an independent Newton's-method fit).
- **A password changed or a name released without the current password**, or a released name's games passing to its next owner: the sign-in check and throttle, ending every session for the name, claim ids in the fit (`TestChangePasswordNeedsTheCurrentOne`, `TestReleaseNameNeedsThePasswordAndFreesTheName`, `TestReleasedClaimLeavesTheRankingAndIsNotIn`, `TestChangePasswordAndReleaseOverHTTP`, `browser_check.py`).
- **A claim acknowledged before it is on disk, or a damaged or escaping data file**: write and flush before acknowledging, torn-tail truncation, refusal of other damage and of symlinks (`TestClaimAndChangeAreNotAcknowledgedUntilDurable`, `TestDataFileIsDurableAndRefusesDamage`, `TestDataPathRefusesSymlinksAndEscapes`).
- **A game that never ends**: passes, the timeout round and the $3N^2$ cap (`TestPassesEndTheGame`, `TestDeadlineTimesOutAndAbandonmentEnds`).

7.4 Derived outputs

Besides the task views, the specification asks for two outputs: the projected-score timeline (per seat, against move number, from bot searches and the viewer's local searches), and the game result (team scores and winners).

Chapter 8

Implementation

`spec/Implementation.lean` maps the specification onto Go. The server (`source/server.go`, `source/game.go`) holds rooms, games and chat in memory behind per-game locks; `source/engine.go` ports the Lab's rules and max-n Monte Carlo search, extended to return the Lab's search summary; `source/bot.go` is the bounded worker pool that realizes the `Bot` actor; `source/channel.go` is the bounded queue whose full sends block until space frees, as the requirements' channel semantics demand.

The browser app is React and TypeScript in `source/web`, so its parts can be rearranged as components: `App.tsx` holds the room tabs, `Room.tsx` a room's Lobby and game tabs, `Game.tsx` one game (board column, play bar and side tabs), `Board.tsx` and `Panels.tsx` the pieces, and `game/game.ts` the controller for one open game (the Lab game tree, server sync, local search). Vite builds it into `source/cmd/demo/static/app/`, which the Go demo embeds. The Lab rules engine, `static/engine.js`, loads first as a plain script and stays the single rules reference for the browser and for `source/engine.go`. The demo serves the engine and search worker as same-origin files, so viewers can branch and search locally under a content security policy that adds only same-origin workers and `blob:`/`data:` images for GIF preview.

The HTTP demo in `source/cmd/demo/` carries the protobuf-shaped messages as bounded JSON. It is an inspection harness for the server core, not evidence of protobuf byte conformance and not a production account system.

8.1 Evidence

- `reports/specification-check/validation.json`: the canonical check of the three specification files, with their hashes.
- `go test -race ./...` in `source/`: unit, concurrency and negative tests named above.
- `reports/browser/`: the two-user Chrome check and its screenshot.
- `book/assets/tour/manifest.json`: the scripted usage tour, with image hashes, source hashes and browser version.

A passing check establishes elaboration, protobuf agreement, structural completeness and the provenance of the figures. It does not prove that the running server satisfies the temporal, isolation, concurrency or resource obligations; those rest on the controls and tests above.

Chapter 9

Usage scenarios

These screenshots come from the running application, driven by a script that plays the parts of three people: alice creates the challenge, bob joins her, and carol watches. `make tour` in the project root reproduces them; `book/assets/tour/manifest.json` records each step, viewer, image hash, source hashes and browser version. Bot moves come from a time-limited search, so the stones differ between runs while the steps do not.

9.1 Welcome

A visitor chooses a local display name. Nothing else is needed for the local preview.

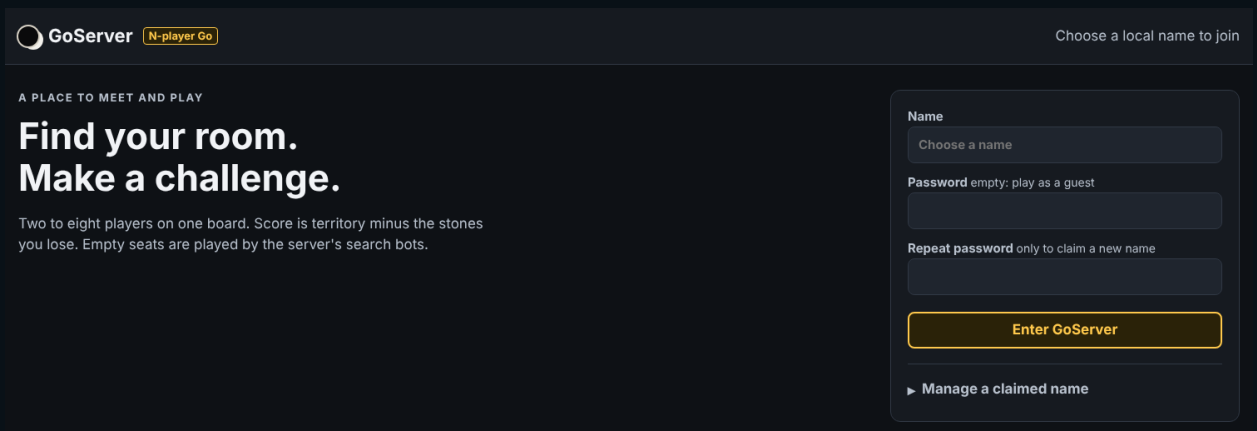


Figure 9.1: Welcome, seen by alice. Steps: open the app.

9.2 Creating a challenge

Alice is in the lobby and fills in a challenge: a 9×9 board, three players, and a byo-yomi clock of one minute of main time and then three periods of ten seconds, the same for people and bots. The ko rule stays on Auto, which is off for three players.

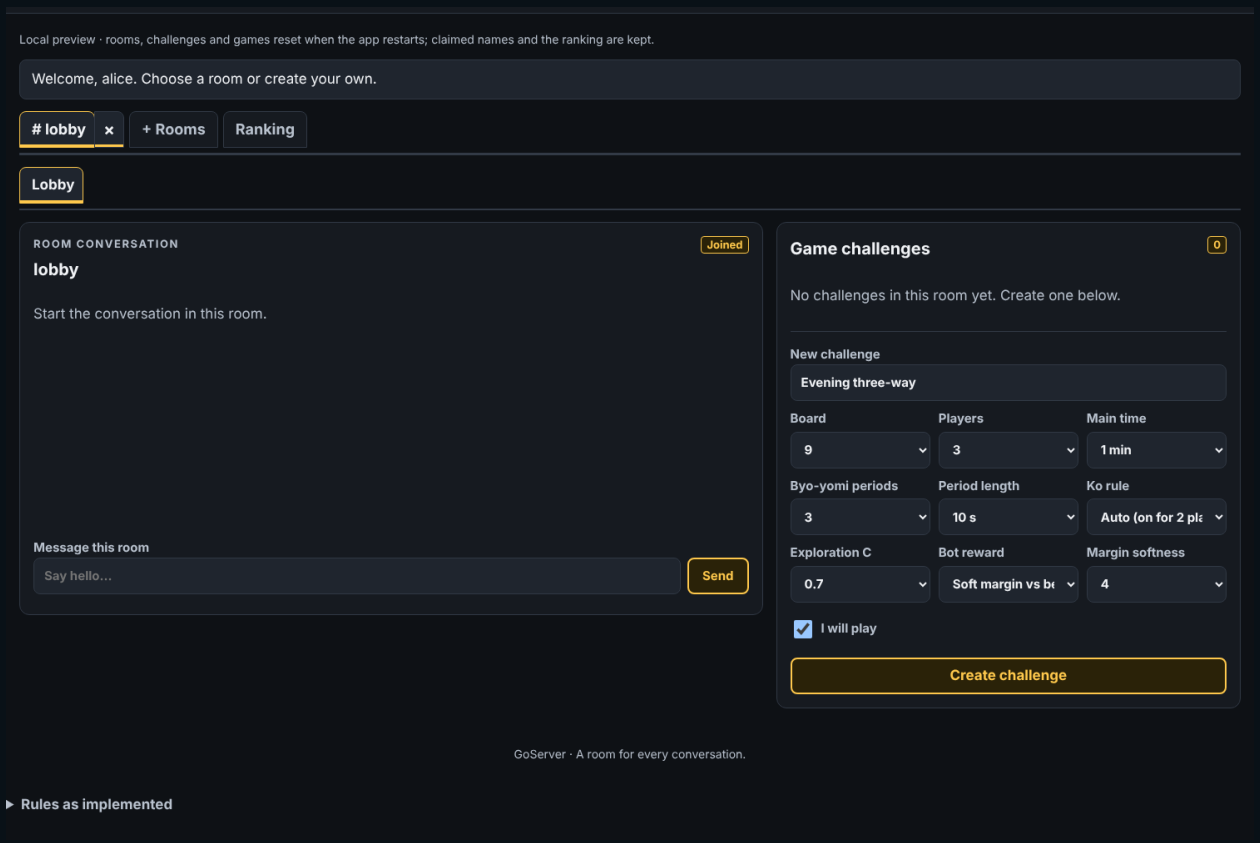


Figure 9.2: Creating a challenge, seen by alice. Steps: sign in as alice; fill in the New challenge form.

9.3 Planning seats

Before the start the creator decides, seat by seat, whether a person or a bot plays and which team the seat is on. Alice took seat 1, bob sat in seat 2, and alice marked seat 3 as a bot.

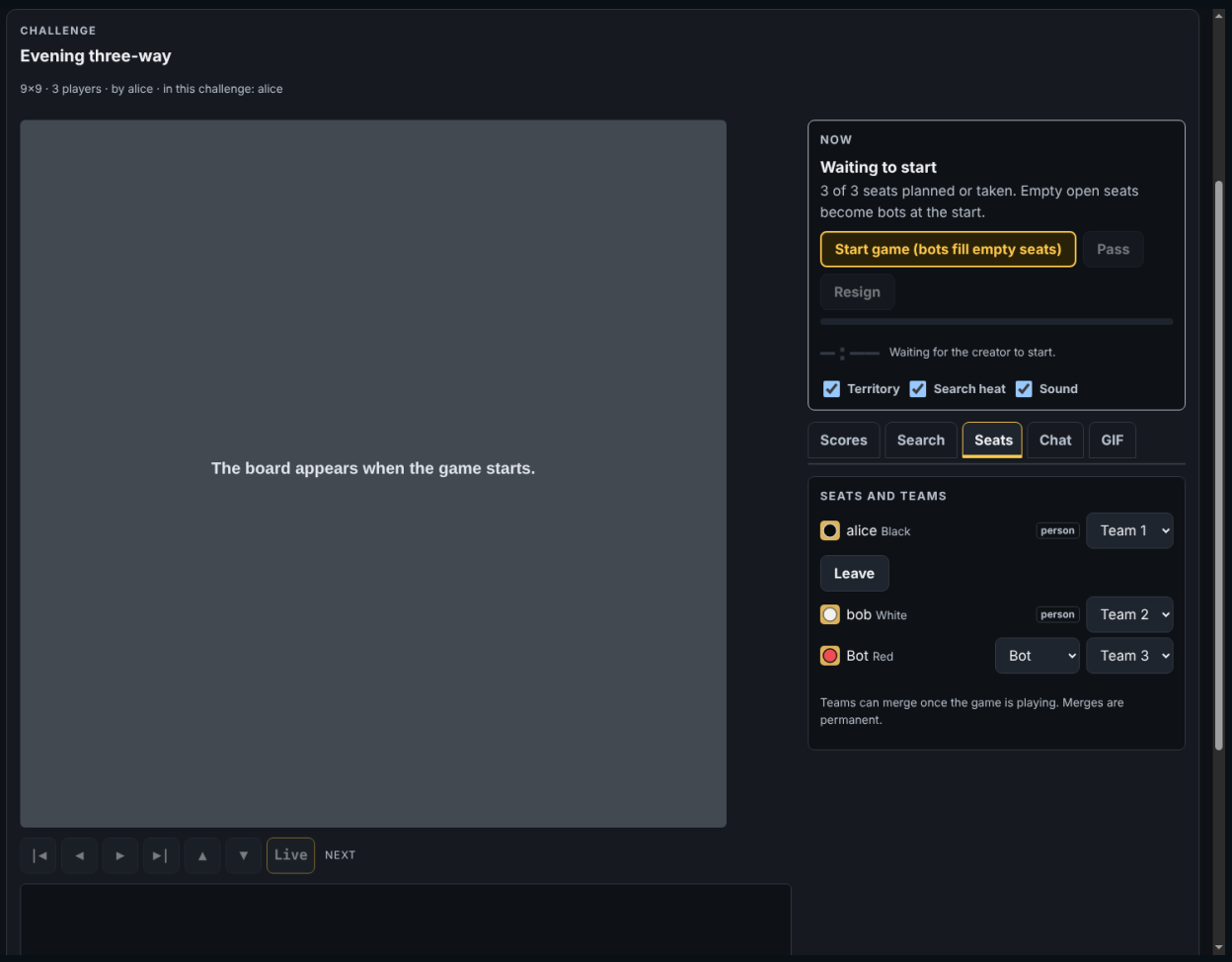


Figure 9.3: Planning seats, seen by alice. Steps: alice: create the challenge (seated automatically); alice: set seat 3 to Bot; bob: open the challenge, Sit here.

9.4 Aiming a stone

On her turn alice moves the pointer over the board: a translucent stone in her colour shows exactly where a click will play. Over an occupied point, a point the ko rule forbids, or when it is not her turn, there is no preview and the pointer shows that a click would be refused. Above the tabs, the play bar keeps whose turn it is and the clock in view.

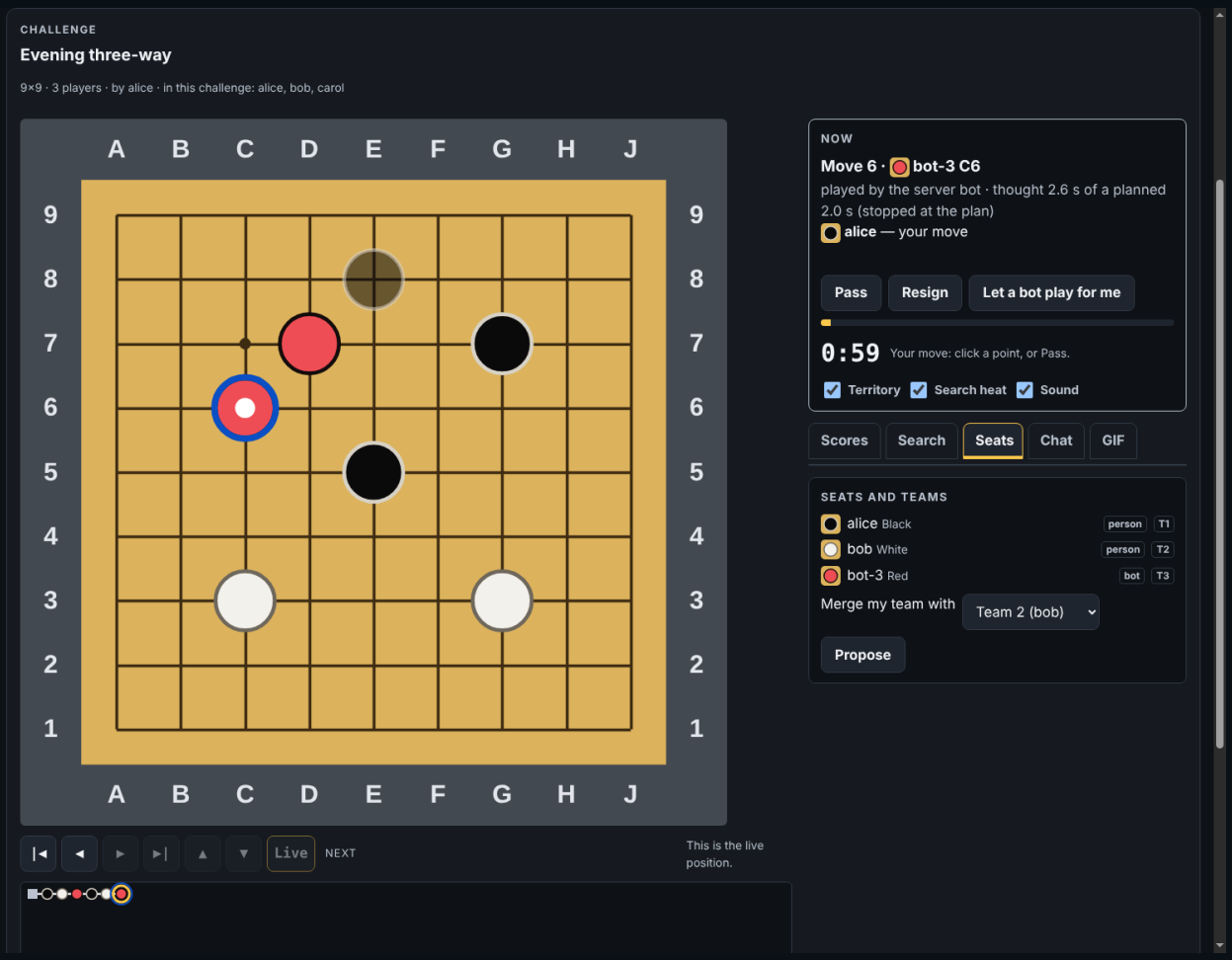


Figure 9.4: Aiming a stone, seen by alice. Steps: alice: move the pointer over an empty point on her turn.

9.5 Reading the bot's search

Alice steps back one move, to the position where the bot searched for its last move. The panel lists the moves that search weighed, with visit share and expected reward; the blue discs on the board are the same search as a heat map. Proj. and Outlook give every seat's projected final score and expected reward, and the timeline under them plots the projections move by move.

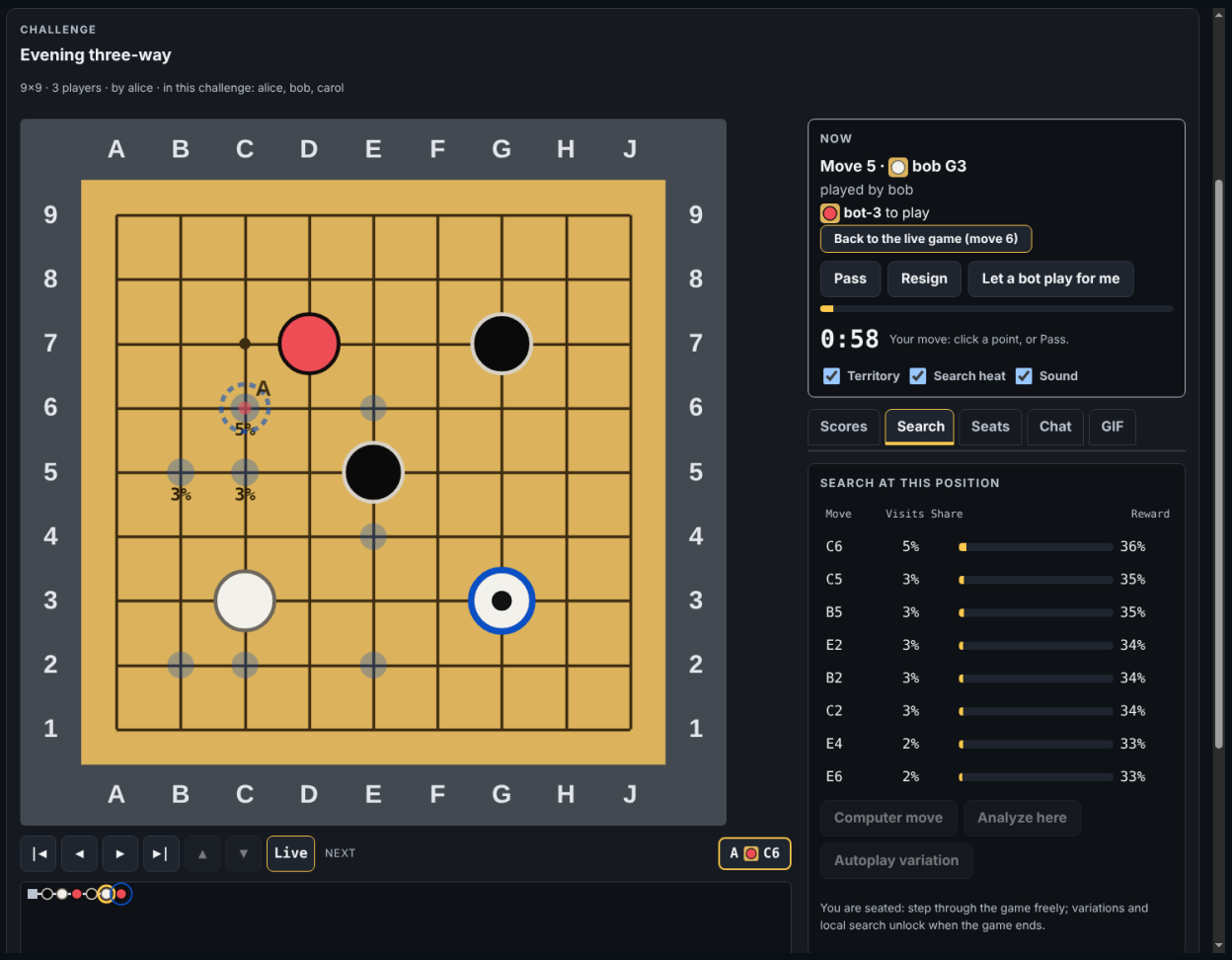


Figure 9.5: Reading the bot's search, seen by alice. Steps: alice and bob alternate stones by clicking the board; the bot answers each round; alice: Back one move.

9.6 A watcher explores a variation

Carol is watching, not playing. She steps back to move 3, tries a different stone (the gold ring marks a variation), and asks for a local search. The analysis runs in her browser only: the dashed branch in the tree never reaches the server, and the live game continues.

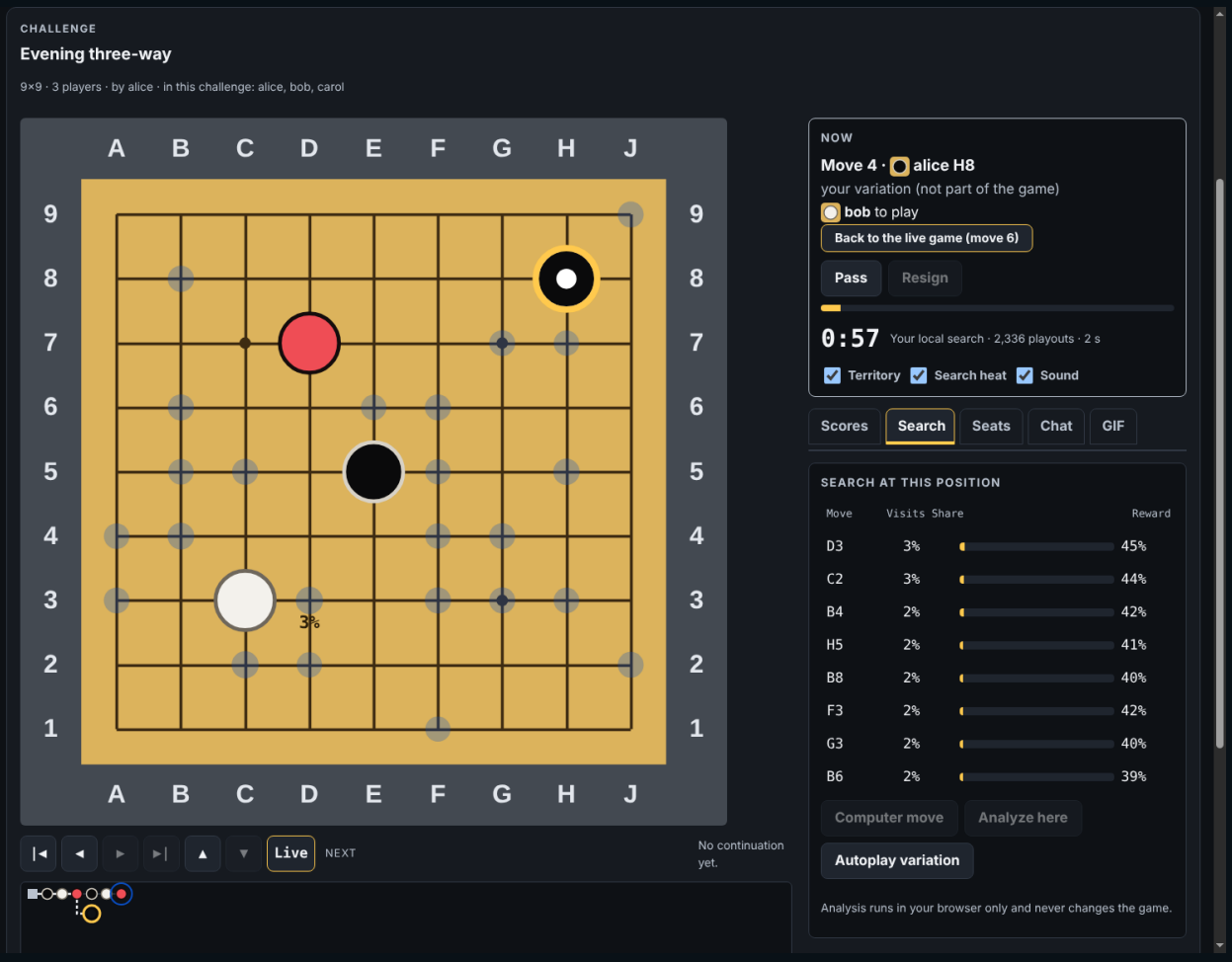


Figure 9.6: A watcher explores a variation, seen by carol. Steps: carol: open the challenge as a watcher; carol: Home, then Forward three times; carol: click an empty point; carol: Analyze here.

9.7 Proposing a team merge

During play bob proposes merging his team with alice's. The Seats tab flags a proposal waiting for her answer; there the proposal shows who has accepted and who is still to answer; it takes effect only when every person on both teams accepts, and it is permanent.

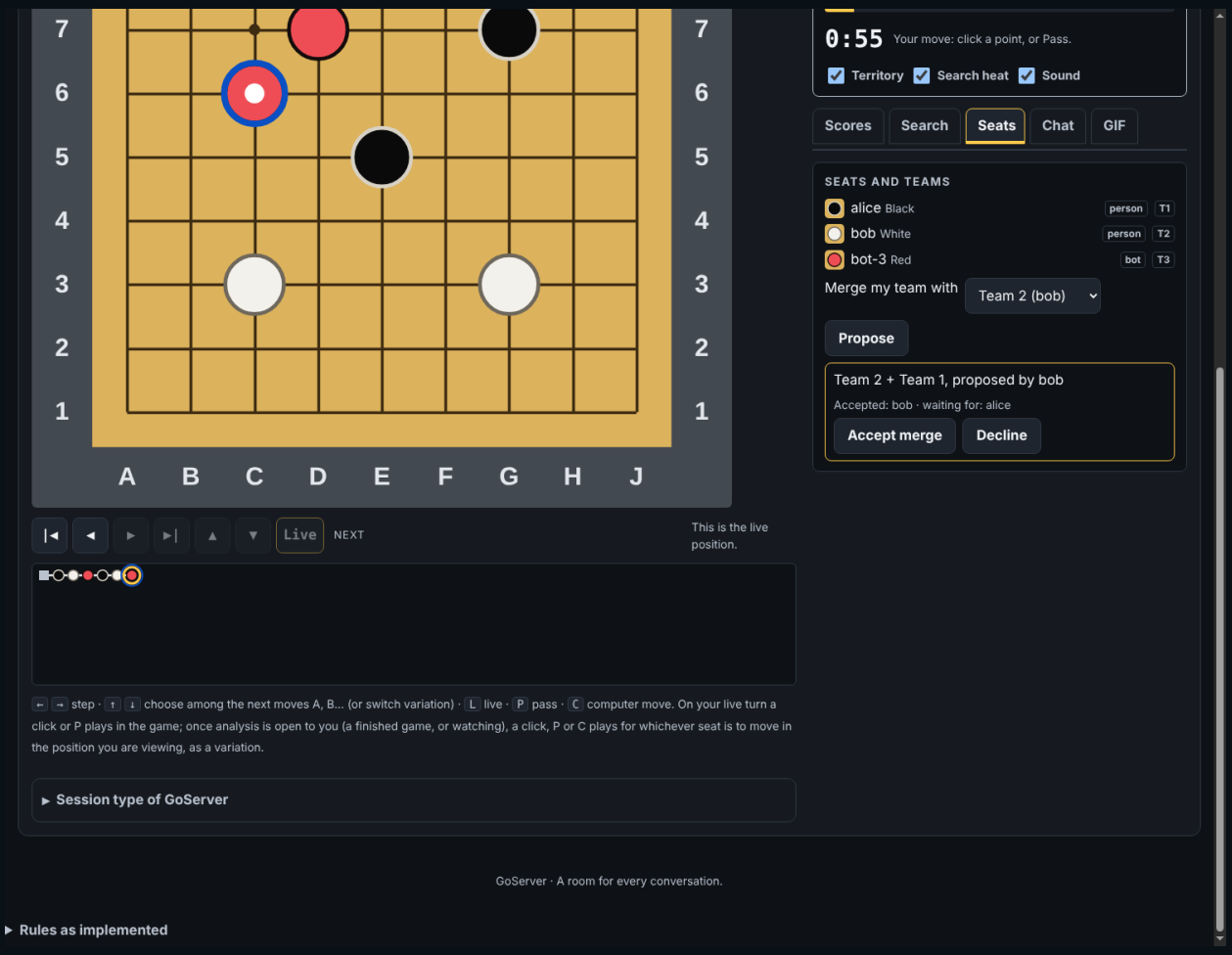


Figure 9.7: Proposing a team merge, seen by alice. Steps: bob: Merge my team with Team 1, Propose.

9.8 Playing as a team, on autopilot

After the merge the score table shows alice and bob as team 1 with a combined total. Alice then lets a bot play her seat; the seat, team and name stay hers, and the bot's move publishes its search like any other bot move. She can take the seat back at any time.

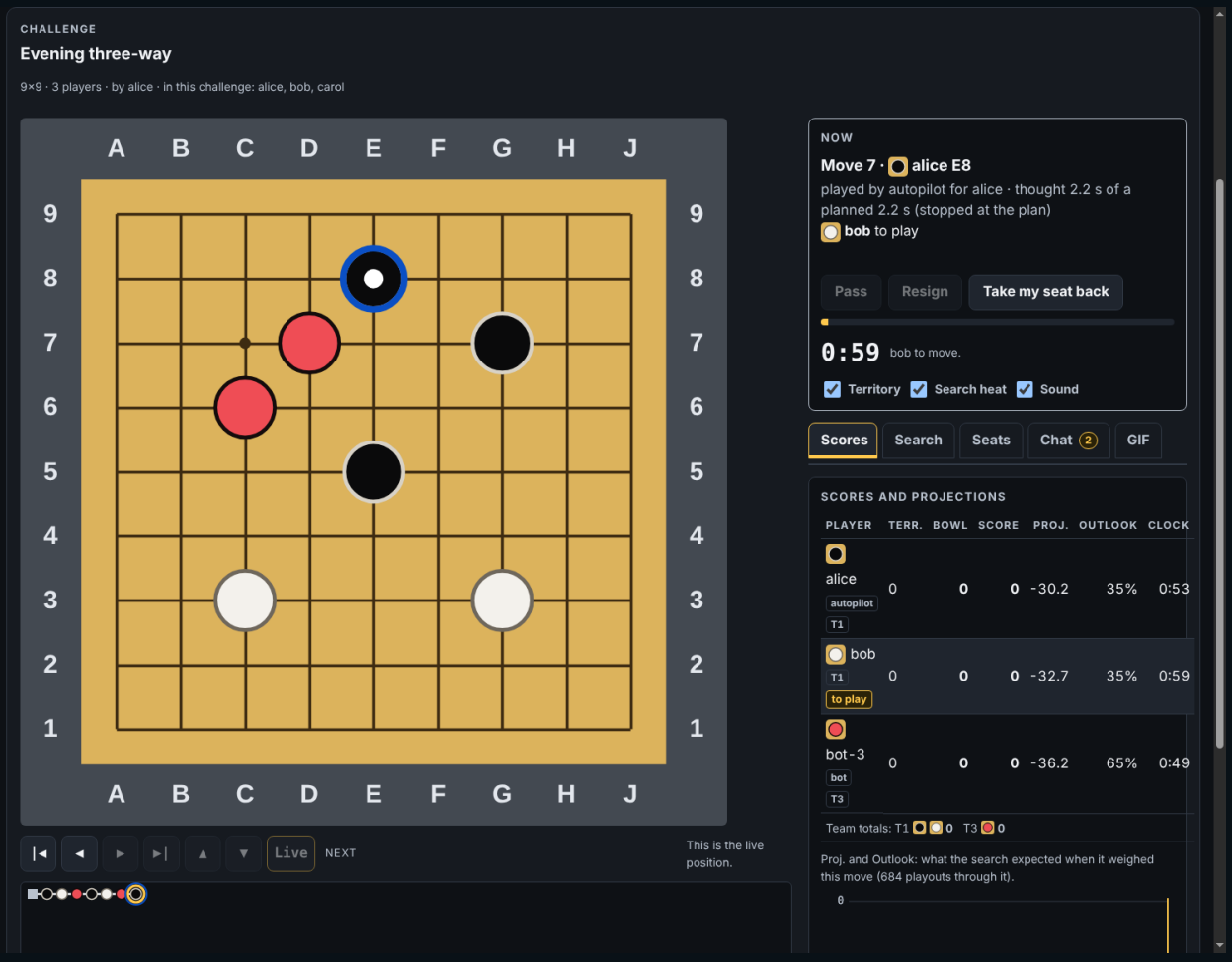


Figure 9.8: Playing as a team, on autopilot, seen by alice. Steps: alice: Accept merge; alice: Let a bot play for me.

9.9 The result

When every remaining player has passed in a row the game ends. The Chat tab lights up with an unread count; the server announces team scores and the winners in the game conversation. Everyone, players included, can now branch and analyse any position.

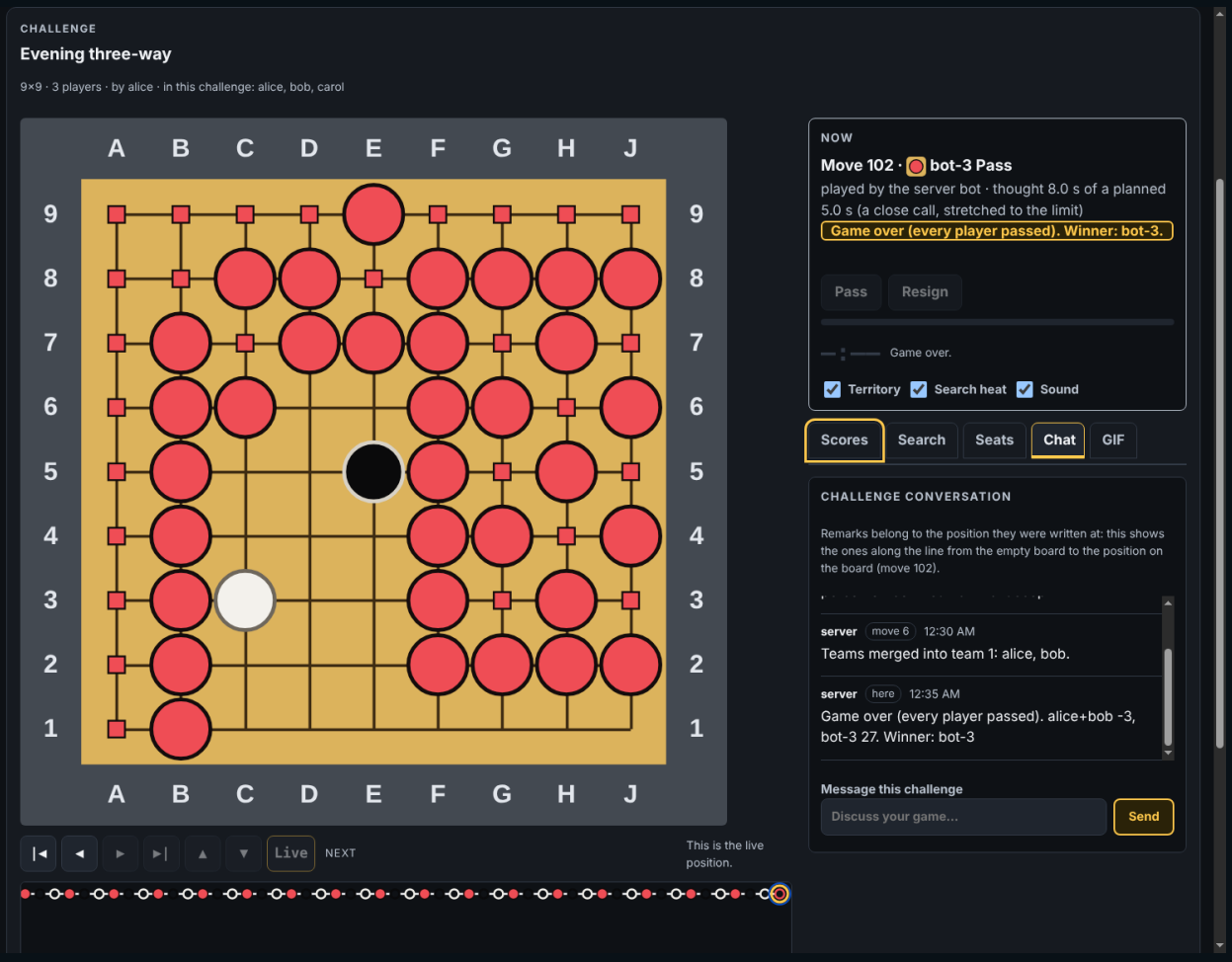


Figure 9.9: The result, seen by alice. Steps: alice passes with the P key, bob with the Pass button, until the bot also passes.

9.10 Reviewing together

After the game alice steps back to move 2, passes for the seat to move there and lets the computer pick the next move. Both moves are kept on the server for everyone in the game, and her remark is pinned to the position she was looking at: the Chat tab shows the remarks along the line on the board, so stepping back hides it again and anyone who walks into this variation sees it.

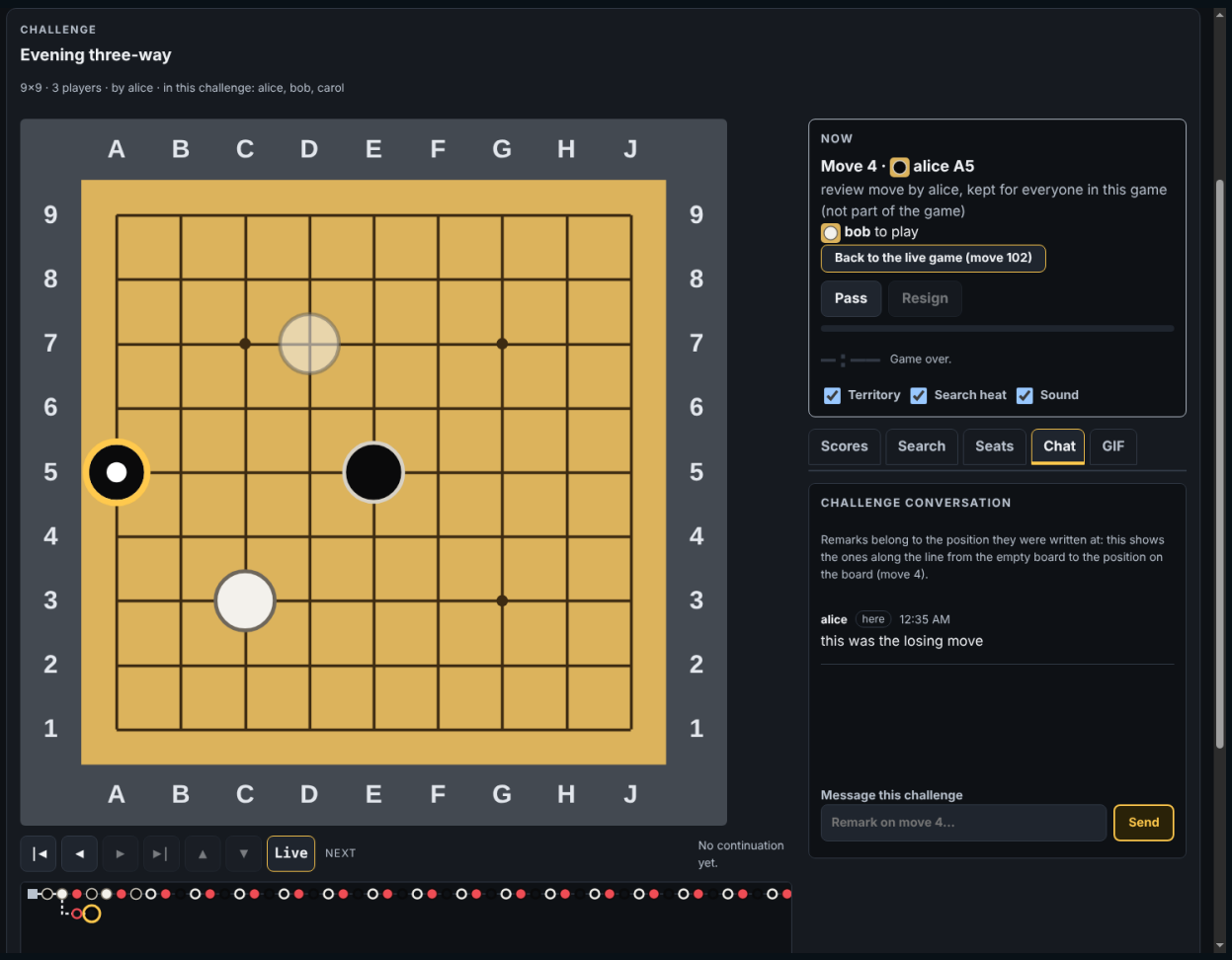


Figure 9.10: Reviewing together, seen by alice. Steps: alice: Home, then → twice; alice: P, then C; alice: write a remark in Chat.

9.11 Rooms and games as tabs

Every room alice has open is a tab along the top, with + Rooms for finding and creating rooms; inside a room, the Lobby (its conversation and challenge list) comes first and every open game has its own tab. Tabs stay alive in the background: a game keeps its variations and analysis, and a tab shows a badge when a move or a merge waits for her or messages arrive. Responses only ever land in the tab that asked for them.

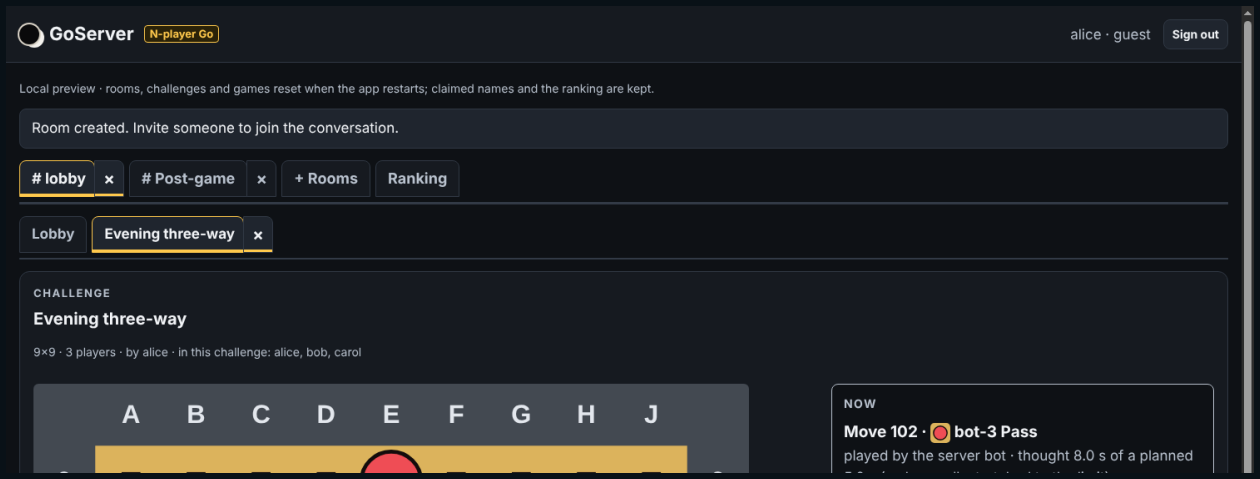


Figure 9.11: Rooms and games as tabs, seen by alice. Steps: alice: + Rooms, create Post-game; alice: back to the lobby room tab.

Chapter 10

Known gaps

This chapter is the work queue. When an item is closed, remove it here and add its evidence under `reports/`.

- Handicaps and tournaments.
- Persistence of rooms, games and chat across restarts, and SGF export of game records (claims and rated results already persist).
- Password recovery; names that look alike in other scripts (homoglyphs) are not treated as the same name.
- Rating uncertainty is not shown; a player with few games simply stays near 1500.
- Push transport: the browser polls (every 0.5 seconds while playing).
- Reconnect grace beyond reloading the page, and moderation.
- Accounts: an unclaimed display name is trusted for the life of the server; anyone may sign in with a claimed name's password, so the password is the account.
- Protobuf byte conformance of the JSON harness.
- Server-side enforcement of the analysis rule for seated players, which is impossible by nature for public positions.

Appendix A

Task catalog

Every task of the checked requirements, as the state machine LeanFM derives from its grammar. Double borders mark terminal states. Regenerate with `make views`.

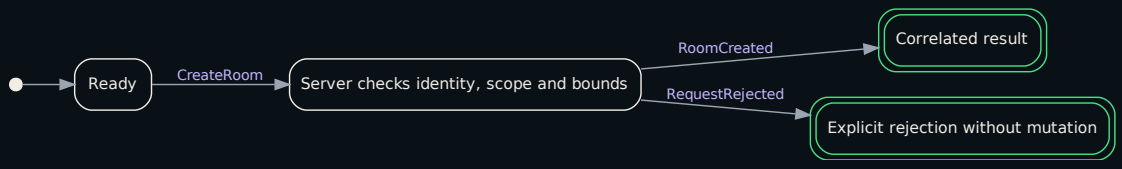


Figure A.1: `create_room`: Create a named chat room

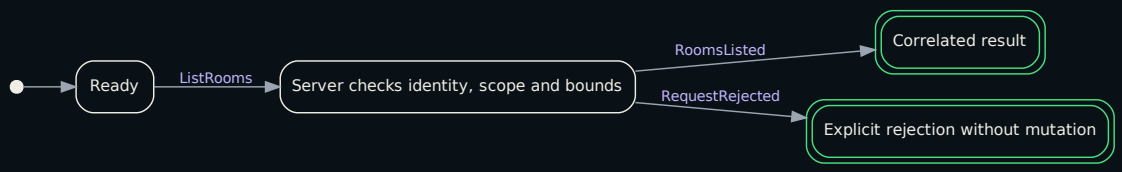


Figure A.2: `list_rooms`: List the available chat rooms

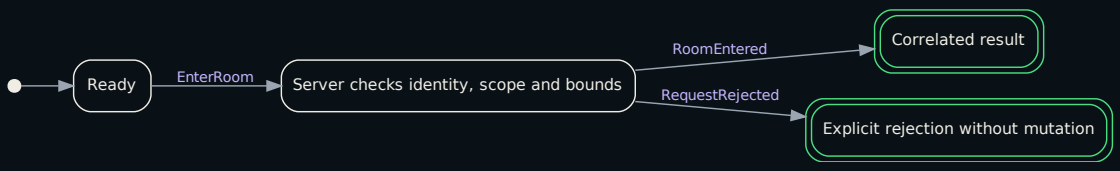


Figure A.3: `enter_room`: Enter a room and inspect its game list

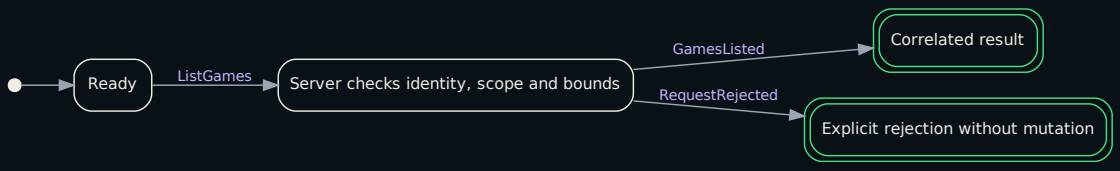


Figure A.4: `list_games`: List room games and open challenges

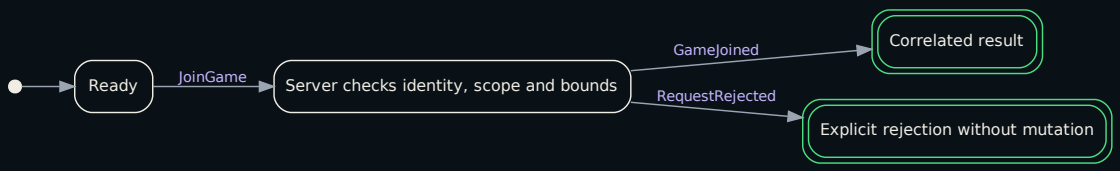


Figure A.5: `join_game`: Watch or enter a game chat container

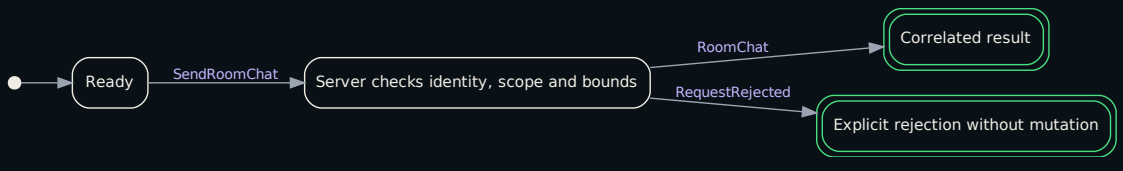


Figure A.6: `room_chat`: Send room chat

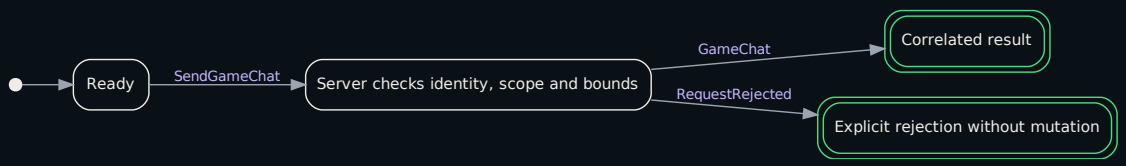


Figure A.7: `game_chat`: Send game chat

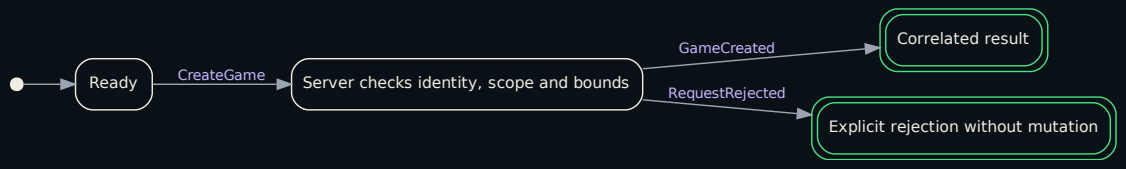


Figure A.8: `create_game`: Create a challenge with board, bot and seat settings

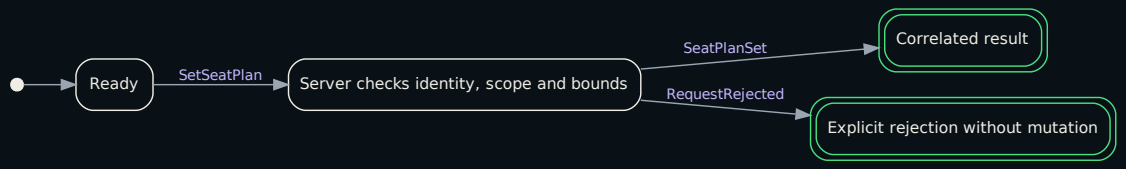


Figure A.9: `set_seat_plan`: Plan a seat as bot or open and set its team

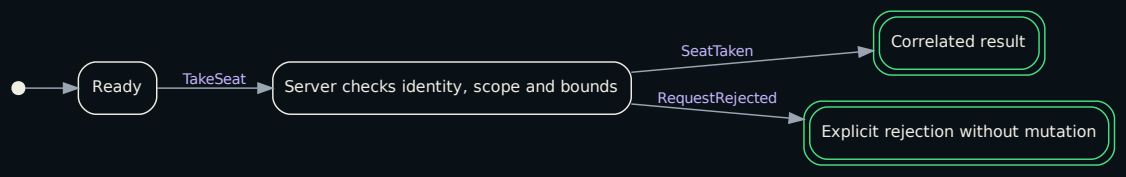


Figure A.10: `take_seat`: Take an open seat

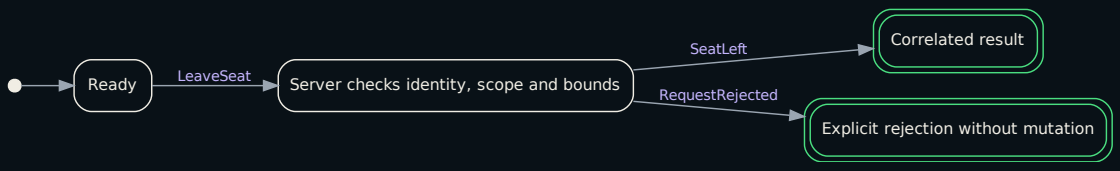


Figure A.11: `leave_seat`: Leave a seat before the start

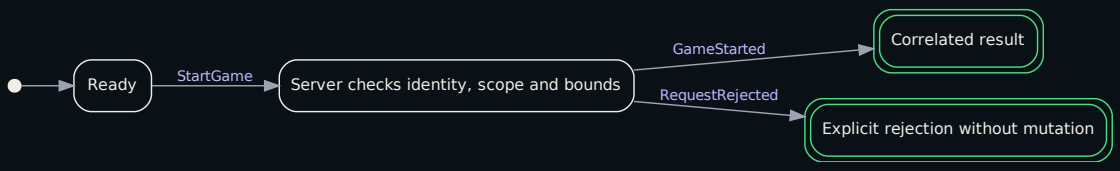


Figure A.12: `start_game`: Start a game; bots fill the remaining seats

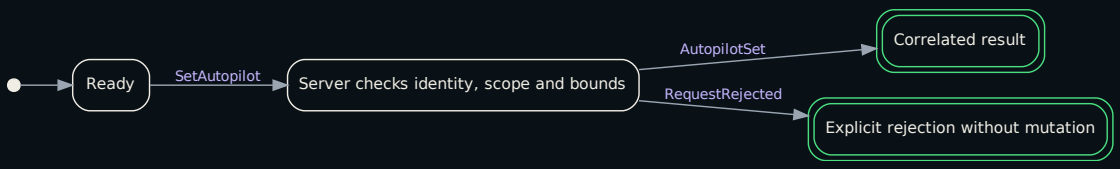


Figure A.13: `set_autopilot`: Hand my seat to a bot, or take it back

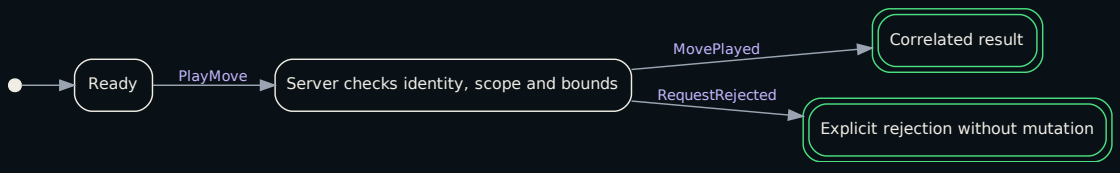


Figure A.14: `play_move`: Play a stone, pass or resign

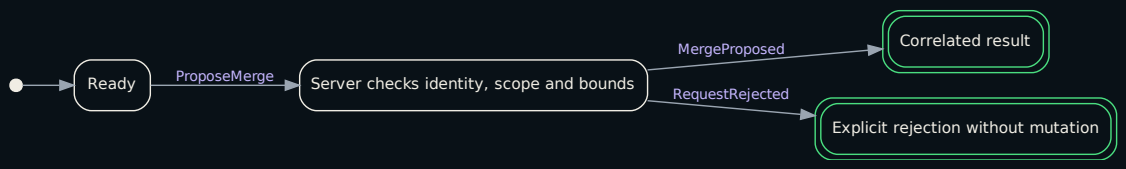


Figure A.15: `propose_merge`: Propose merging two teams

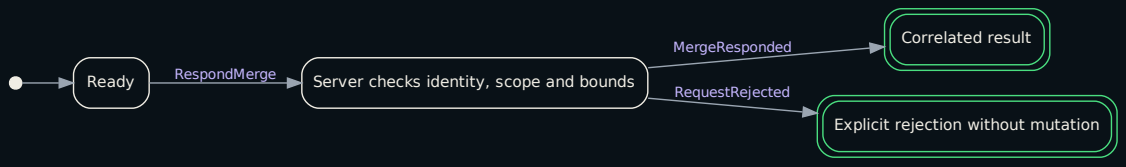


Figure A.16: `respond_merge`: Accept or decline a merge proposal

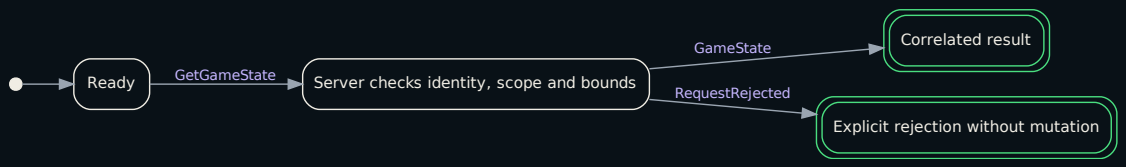


Figure A.17: `game_state`: Watch the board, clocks, scores, teams and record

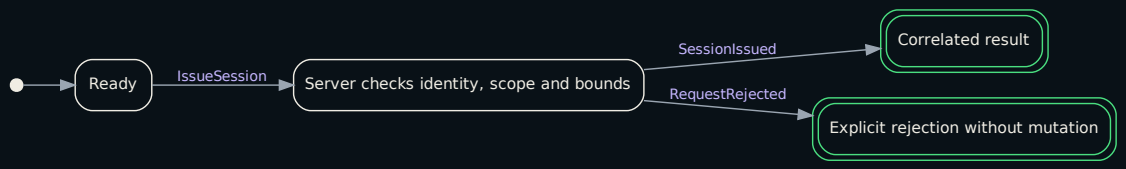


Figure A.18: `sign_in`: Enter a name, with its password if it is claimed

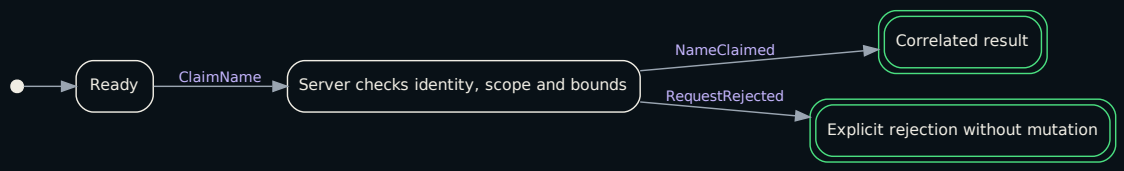


Figure A.19: `claim_name`: Claim a name: enter it with a password typed twice

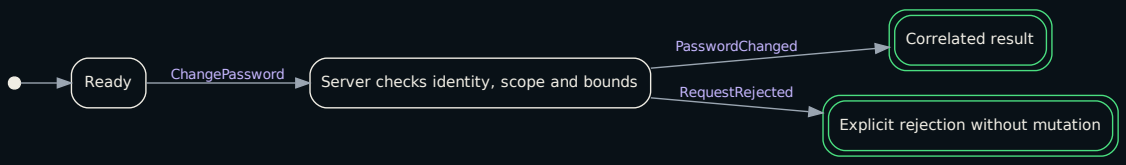


Figure A.20: `change_password`: Change my password: the old one once, the new one twice

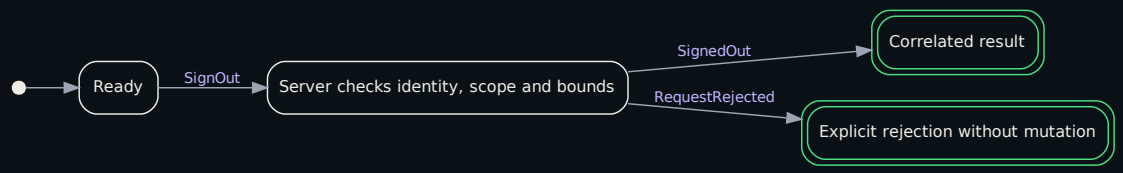


Figure A.21: `sign_out`: Sign out and return to the sign-in form

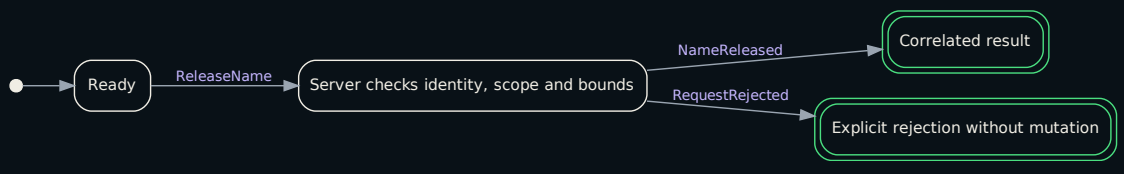


Figure A.22: `release_name`: Release my claimed name

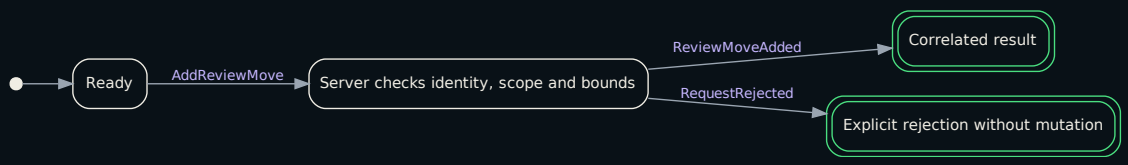


Figure A.23: `review_move`: Add a move to a finished game's shared review tree

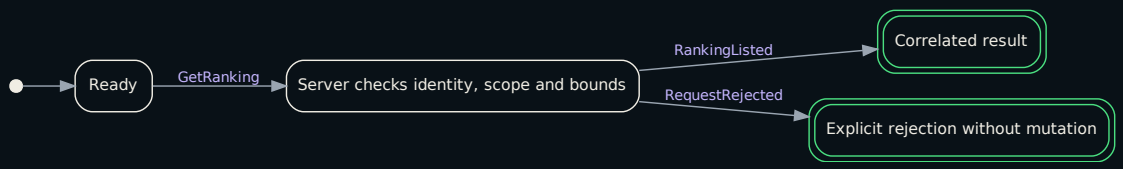


Figure A.24: `ranking`: See the ranking of claimed names and bots

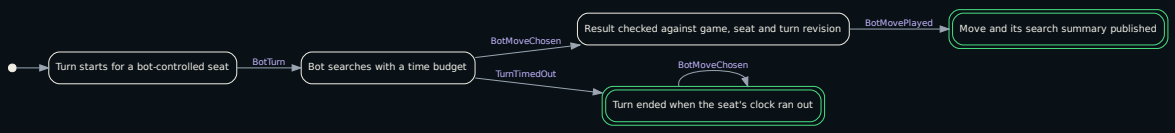


Figure A.25: `bot_turn`: A server bot plays one turn

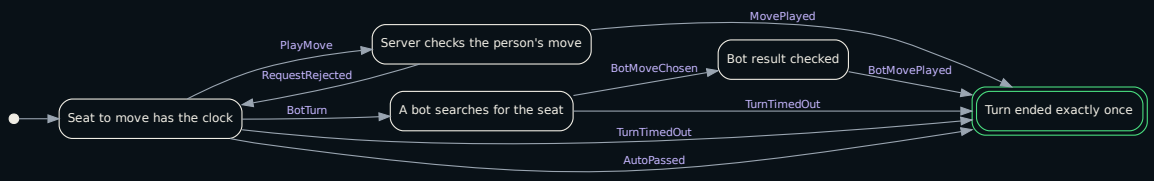


Figure A.26: `turn`: One turn of a started game

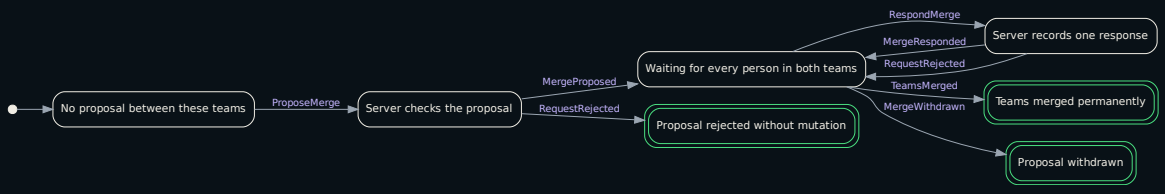


Figure A.27: `team_merge`: Two teams merge by consent



Figure A.28: `game_lifecycle`: A game from challenge to result

Appendix B

How the project got here

The git history is the authoritative log. In outline: the first slice was a server-only room, challenge and chat service with typed correlation and replay; working room creation and a real browser followed; then N-player play with server bots, deadlines and scoring; then parity with the N-Player Go Lab (seat plans, autopilot, teams and merges, search summaries, history, variations and local analysis). On November 1, 2026 the project moved to the repository's current conventions: review in git instead of recorded approval gates, a project-level Makefile, figures derived from the checked catalog, and this book rewritten to describe the system as it is.